THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 1 & 2: **Patterns in Mathematics, Lines and Angles** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 216	the rule gives 216
2	**(a)** 243	the rule gives 243
3	**(a)** 36	sum of first n odd numbers is n^2
4	**(b)** 16	a square number
5	**(d)** 6	$3 + 2 + 1$
6	**(a)** endlessly in both directions	no end points
7	**(c)** intersecting lines	meet
8	**(a)** degree	symbol °
9	**(a)** 180°	half turn
10	**(c)** 180°	straight line

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Add 3 each time; next term is 14.	1+1
12	$1 + 2 + 3 + 4 + 5 + 6 = 21$ dots.	1+1
13	Two lines that meet at a right angle.	1+1
14	90°; right angle.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Each term doubles the previous one (1); 1, 2, 4, 8, 16, 32, 64, 128 (2).	1+2
16	180° (1); 70° (2).	1+2
17	Any valid rule with five correct terms, e.g. multiply by 3: 1, 3, 9, 27, 81.	1+2
	OR: A regular polygon has 3, 4, 5, … sides (1.5); the sides increase by one each time just like counting numbers from 3 (1.5).	1½+1½
18	A full turn of 360° (1.5); 4 right angles (1.5).	1½+1½
	OR: 20°, 90°, 135°, 180°, 270° (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) ₹32 (1). (ii) Powers of 2 (1). (iii) ₹31 (1). (iv) Day 8 (₹128) (2).	1+1+1+2
	OR: (i) Two triangles form a 3 by 3 square (2). (ii) 25 (1.5). (iii) 100 (1.5).	2+1½+1½
20	(i) 360° (1). (ii) 140° (2). (iii) Obtuse (2).	1+2+2
	OR: (i) Acute, right, obtuse, straight (2). (ii) 3/8 (1). (iii) 8 (1). (iv) 90° (1).	2+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 1 & 2: **Patterns in Mathematics, Lines and Angles** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 3	common difference
2	**(b)** 99	$2 \times 50 - 1$
3	**(b)** 10	$1 + 2 + \cdots + 4 = 10$
4	**(c)** 78	$1 + 2 + \cdots + 12 = 78$
5	**(b)** 400	$n^2 = 400$
6	**(c)** 20°	comparison
7	**(d)** arms	sides of the angle
8	**(a)** acute	compare with 90°, 180°, 360°
9	**(d)** reflex	compare with 90°, 180°, 360°
10	**(c)** obtuse	compare with 90°, 180°, 360°

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	24 and 23.	1+1
12	Square numbers: 4, 9, 16, …	1+1
13	A straight angle by 90°.	1+1
14	Trace one angle and place it over the other so vertices and one arm coincide; compare the other arms.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	6, 12, 18 (1.5); the differences are multiples of 6 (1.5).	1½+1½
16	Fix two strips with a pin (1.5); rotate one strip to form angles of different measures (1.5).	1½+1½
17	Patterns help make predictions (1); explain nature and science (1); lead to new ideas and theories (1).	1+1+1
	OR: 16 (1); adding counting numbers up and down gives square numbers (2).	1+2
18	At 2:00 it is 60° (1.5); at 10:00 it is 60° (1.5).	1½+1½
	OR: Acute is less than 90°, e.g. 40° (1.5); obtuse is between 90° and 180°, e.g. 130° (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Correct table (3). (ii) Both increase by 2; odd numbers are one less than even (2).	3+2
	OR: (i) Dots arranged in hexagons (1). (ii) 1, 7, 19, 37, 61 (2). (iii) 6, 12, 18, 24 multiples of 6 (2).	1+2+2
20	(i) Standard unit (1). (ii) Historical convention dividing a circle into 360 parts (2). (iii) 60° (2).	1+2+2
	OR: (i) Three accurate angles (3). (ii) Acute, right, obtuse (1). (iii) 150°, 90°, 40° (1).	3+1+1

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ANSWER KEY & MARKING SCHEME SET C

Chapters 1 & 2: **Patterns in Mathematics, Lines and Angles** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 32	the rule gives 32
2	**(c)** 55	$1 + 2 + \cdots + 10 = 55$
3	**(a)** 81	9^2
4	**(d)** the sum of the two terms before it	rule of the sequence
5	**(d)** 10	$4 + 3 + 2 + 1$
6	**(a)** one starting point and extends endlessly in one direction	definition
7	**(d)** two rays with a common starting point	vertex and arms
8	**(d)** right angle	quarter turn
9	**(c)** reflex angle	between 180° and 360°
10	**(a)** 30°	$360 \div 12$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$5 + 4 + 3 + 2 + 1 = 15$ lines.	1+1
12	Calendars, designs on floors and cloth, music and nature (any two).	1+1
13	An angle between 180° and 360°; e.g. 250°.	1+1
14	4 right angles.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Pictures make the relation visible (1.5); example of dots in squares for 1, 4, 9 (1.5).	1½+1½
16	270° (1.5); reflex (1.5).	1½+1½
17	1, 2, 3, 5, 8, 13, 21, 34, 55 (2); 34 (1).	2+1
	OR: $7^2 = 49$, $5^3 = 125$ (2); so the 5th cube is greater (1).	2+1
18	90°, 180° and 270° (3).	3
	OR: Lines that cross each other (1.5); exactly one common point (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Add 2 each time (1). (ii) 21 (1). (iii) Figure 20 (1). (iv) Odd numbers from 3 (2).	1+1+1+2
	OR: (i) 6 by 6 square (2). (ii) 400 (1.5). (iii) Yes; 13 (1.5).	2+1½+1½
20	(i) Points on a line (1). (ii) Exactly one (2). (iii) Three (2).	1+2+2
	OR: (i) $4 \times 30 = 120^\circ$ (2.5). (ii) $40 \times 6 = 240^\circ$ (2.5).	2½+2½

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ANSWER KEY & MARKING SCHEME SET D

Chapters 1 & 2: **Patterns in Mathematics, Lines and Angles** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 80	doubling
2	**(a)** dots arranged in hexagons around a centre	centred shape
3	**(c)** 21	$1+2+\cdots+6=21$
4	**(c)** 49	$n^2=49$
5	**(a)** 625	$n^2=625$
6	**(d)** 90°	perpendicular directions
7	**(a)** 180	half of 360
8	**(c)** acute	compare with 90°, 180°, 360°
9	**(c)** reflex	compare with 90°, 180°, 360°
10	**(d)** reflex	compare with 90°, 180°, 360°

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Yes; $10\times10=100$.	1+1
12	36, a square number.	1+1
13	Place its centre on the vertex, base line along one arm and read from the zero of the scale.	1+1
14	Corners of a book, scissors blades, clock hands (any two).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Start with a triangle (1); replace each line segment by a speed bump (1); repeat and get a snowflake-like shape (1).	1+1+1
16	Estimate compared with 90° (1.5); measure with protractor (1.5).	1½+1½
17	29 (1); $15^2=225$ (2).	1+2
	OR: 1234321 (2); pattern rises to the number of ones then falls (1).	2+1
18	30° acute, 90° right, 150° obtuse (3).	3
	OR: Angle measures the turn between arms, not their lengths (3).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Any valid situation with correct six terms and a stated rule.	2+2+1
	OR: (i) Sequences and shapes follow rules (2). (ii) Petals, honeycomb, calendar, rangoli (2). (iii) Predicts and explains (1).	2+2+1
20	(i) Fold paper twice to make a corner (2). (ii) Fold the right angle to bisect it for 45° (3).	2+3
	OR: (i) So wheels keep a steady distance (2). (ii) Crossroads, scissors (3).	2+3

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.